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Quantum Fisher information and spin squeezing in the ground state of the XY model

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Abstract
We analyze the quantum Fisher information (QFI) and entanglement of the ground state for the XY model. The QFI determines the precision in parameter estimations, moreover, it is a criterion of entanglement. The scaling behaviors of the QFI are studied both at the critical point and in different phases. In the symmetry-broken phase, the QFI scales proportional to the square of the system size, while in the symmetric phase, it is nearly independent of the system size. The study of QFI not only reveals the distinct entanglement properties of the XY model in different phases, but also shows that the ground state of the XY model can be used to perform Heisenberg-limit parameter estimation.

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(Some figures may appear in colour only in the online journal)

1. Introduction

A quantum phase transition [1] is driven by the change of the external parameters at absolute zero temperature. Conventionally, second-order quantum phase transitions are characterized by an order parameter. In the last decade, it was shown that quantum phase transitions can also be detected by entanglement [2–7] and fidelity susceptibility [8–17]. Intensive studies of entanglement in various phase transition systems have achieved much success [7]. When quantum phase transitions occur, the symmetry of the ground state changes dramatically and the correlation length is divergent. Take the XY model for example, the pairwise entanglement [2, 3] and the block–block entanglement [4–6] have been studied by using concurrence [18] and entanglement entropy, respectively. Both quantities exhibit singular behaviors near the critical point. However, each entanglement measure has its limitation, and the rich phenomena of entanglement in interacting spin systems needs to be explored from different points of view.

In this work, we use a recently proposed entanglement criterion, the quantum Fisher information (QFI), to study the quantum phase transition. QFI is a very basic concept in parameter estimation theory [19]. Consider the problem of estimating a parameter θ from a quantum state $\rho(\theta)$. By measuring a proper physical observable, the value of θ can be estimated from the measurement results. The precision of the estimation of θ, which is limited by unavoidable measurement errors, is determined by the QFI (see equation (4)). Recently, an interesting proof showed that [20] QFI not only gives the limitation in parameter estimation, it can also detect entanglement. In fact, before this proof, several experiments have already been performed to demonstrate the improvement of phase estimation precision by using entangled states [21, 22].

To study quantum phase transitions, we choose the one-dimensional XY model. The choice is based on the following considerations. (i) The XY model can be solved analytically. The phase transition of the XY model is clear enough and has been studied by various approaches. It will offer a benchmark to test the QFI in the proximity of quantum phase transitions. (ii) The behaviors of QFI in phase transitions were studied in the Lipkin–Meshkov–Glick (LMG) model [23, 24]. The LMG model can be viewed as an XY model with infinite range interactions. Since both models have second-order quantum phase transitions, self-consistent results are expected to be obtained to examine the ability of QFI in detecting phase transitions. (iii) The XY model could be realized in some artificial systems, such as Josephson junction arrays [25] and neutral atoms in optical lattice [26–28].

We find that the entanglement of the ground state for the XY model in different quantum phases can be distinguished very well. Moreover, in the symmetry-broken phase, the entanglement of the ground state could be a potential resource in parameter estimations. We also compute the spin squeezing parameter [29–32] in order to make a comparison with the QFI. We find that the spin squeezing parameter is nearly independent of the system size, which is different from that in the LMG model [33]. At the critical point, the derivative of the spin squeezing parameter with respect to the strength of the transverse field is also divergent in the thermodynamic limit.

Our work is organized as follows. In section 2, we give a brief introduction of the parameter estimation theory, including the definitions and physical signatures of QFI and spin squeezing parameters. Then in section 3, we study the QFI and spin squeezing parameter of the ground state for the XY model. Finally, a conclusion is given in section 4.

2. QFI and the spin squeezing

2.1. Maximal QFI

In this section we give a brief introduction of parameter estimation theory and QFI. The QFI is the information of a parameter that we can get from a quantum state. It is related to the degrees of statistical distinguishability of a quantum state from its neighbors in parameter space. Consider a quantum state $\rho\,(\theta)$, θ is the parameter that we need to estimate. The QFI of θ is defined as

$$F_Q(\theta) = \mathrm{tr}\big[\rho(\theta)\mathrm{L}_\theta^2\big], \tag{1}$$

where L_θ is the so-called symmetry logarithmic derivative, determined by the following equation [19]

$$\partial_\theta \rho(\theta) = \tfrac{1}{2}[L_\theta \rho(\theta) + \rho(\theta) L_\theta], \tag{2}$$

where $\partial_\theta := \partial/\partial\theta$, and L_θ can be resolved by rewriting the above equation under the eigenbasis of ρ. Then the QFI can be obtained as

$$F_Q(\theta) = \sum_i \frac{(\partial_\theta p_i)^2}{p_i} + 2\sum_{i\neq j} \frac{(p_i - p_j)^2}{p_i + p_j} |\langle\varphi_j|\partial_\theta\varphi_i\rangle|^2, \tag{3}$$

where $|\varphi_i\rangle$ and p_i are eigenvectors and eigenvalues of ρ, respectively. If we want to estimate the parameter θ through measuring $\rho(\theta)$, the variance of our estimation is limited by a fundamental bound, given by the quantum Cramer–Rao (QCR) [19] inequality

$$(\Delta\theta)^2 \geqslant (\Delta\theta_{\rm QCR})^2 \equiv \frac{1}{F_Q(\theta)}. \tag{4}$$

If $F_Q(\theta)$ is large, the variance of our estimation is small.

The above discussion of the QFI is for arbitrary quantum states and parameters. In our work, since the XY model is a spin model, the quantum state is limited in a specific class, hence we shall give a more concrete discussion of the QFI. Consider the following scenario, which can be seen in general quantum metrology process. Firstly, we prepare a system consisting of N spin-1/2 particles, which is treated as a probe. Then it is rotated around the direction $\vec{n}$ for a certain angle θ. This process (quantum channel) is formulated as $\rho(\theta) = e^{i\theta S_{\vec{n}}}\rho_{\rm in}\, e^{-i\theta S_{\vec{n}}}$. Finally, we estimate the angle θ by measuring $\rho(\theta)$. The variance of our estimation is bounded by the QCR. The QFI for θ is denoted as $F_Q[\rho_{\rm in}, S_{\vec{n}}]$. As the eigenvalues of $\rho(\theta)$ and $\rho_{\rm in}$ are the same, the expression of QFI (7) is just [20, 34, 35]

$$F_Q[\rho_{\rm in}, S_{\vec{n}}] = 2\sum_{i,j} \frac{(p_i - p_j)^2}{p_i + p_j} |\langle\psi_j|S_{\vec{n}}|\psi_i\rangle|^2. \tag{5}$$

For a pure state $\rho_{\rm in} = |\psi\rangle\langle\psi|$, the QFI is simplified as

$$F_Q[\rho_{\rm in}, S_{\vec{n}}] = 4(\Delta S_{\vec{n}})^2 \tag{6}$$

where $(\Delta S_{\vec{n}})^2 \equiv \langle S_{\vec{n}}^2\rangle - \langle S_{\vec{n}}\rangle^2$.

In our work, the QFI is actually studied via the following parameter

$$\chi^2 = \frac{N}{F_Q[\rho_{\rm in}, S_{\vec{n}}]}, \tag{7}$$

which was proposed recently by Pezzé and Smerzi [20], who also proved that $\chi^2 < 1$ is a sufficient condition for multipartite entanglement. The following relation

$$(\Delta\theta)^2 \geqslant \frac{\chi^2}{N} \tag{8}$$

gives a connection between $(\Delta\theta)^2$ and χ^2. If $\chi^2 = 1$, the lower bound of $\Delta\theta$ is $1/N$, which is called the shot-noise estimation. If $\chi^2 < 1$, the state is entangled, and we can attain the sub-shot-noise limit. When $\chi^2 = 1/N$, the estimation of $(\Delta\theta_{\rm QCR})^2 = 1/N^2$, i.e. the Heisenberg limit [36], which is the highest precision in parameter estimation. Below, we only use equation (6), since the ground state of the XY model is a pure state.

We note that the above QFI depends on direction $\vec{n}$, thus it is not definite for a given state. In practice, we hope to find a direction $\vec{n}$, of which the corresponding QFI is maximal and χ^2 is minimal. It was shown in [37] that the maximal QFI is

$$\max_{\vec{n}} F_Q[\rho_{\rm in}, S_{\vec{n}}] = \lambda_{\max}, \tag{9}$$

where $\lambda_{\max}$ is the largest eigenvalue of matrix $\mathbf{C}$, of which the matrix elements are given by

$$\mathbf{C}_{kl} = \sum_{i\neq j} \frac{(p_i - p_j)^2}{p_i + p_j} [\langle\varphi_i|S_k|\varphi_j\rangle\langle\varphi_j|S_l|\varphi_i\rangle + \text{h.c.}], \tag{10}$$

where $k, l \in x, y, z$. For pure states, the above matrix $\mathbf{C}$ can be simplified as

$$\mathbf{C}_{kl} = 2\langle S_k S_l + S_l S_k \rangle - 4\langle S_k \rangle \langle S_l \rangle, \tag{11}$$

then $\mathbf{C}$ is the covariance matrix. In the next section, we shall discuss spin squeezing parameters, as well as their relations with χ^2.

2.2. Spin squeezing parameters

The most studied spin squeezing parameters are [29, 32]

$$\xi_S^2 = \frac{4 \min(\Delta S_{\vec{n}_\perp})^2}{N}, \quad \xi_R^2 = \frac{N \min(\Delta S_{\vec{n}_\perp})^2}{|\langle S_{\vec{n}} \rangle|^2}, \tag{12}$$

where the subscript $\vec{n}_\perp$ refers to an arbitrary axis perpendicular to the mean-spin direction $\vec{n} = \langle \vec{S} \rangle / |\langle \vec{S} \rangle|$. It was proved that spin squeezing parameters are criteria for multipartite entanglement [38, 39]. If $\xi_i^2 < 1$ $(i = S, R)$, the state is entangled. Spin squeezing and QFI have close relations. (i) Spin squeezed states can be used to reduce the measurement uncertainty and improve the precision of the atomic clock [40, 41]. It plays a similar role as QFI in parameter estimations. (ii) Both $\chi^2 < 1$ and $\xi_R^2 < 1$ are sufficient conditions for multipartite entanglement. However, as proved in [20], $\chi^2 \leqslant \xi_R^2$, and thus a state could be entangled ($\chi^2 < 1$) but not spin squeezed ($\xi_R^2 \geqslant 1$), such as the Dicke states [23]. In the next section we will see that, although spin squeezing and QFI have close relations, they behave quite different in the XY model.

3. QFI and spin squeezing in the ground state of the XY model

3.1. XY Hamiltonian

In this section, we shall study the behaviors of χ^2 and ξ_S^2 in the XY model. The Hamiltonian of the ferromagnetic XY model is

$$H(\gamma, \lambda) = -\sum_{i=-M}^{M} \left(\frac{1+\gamma}{2} \sigma_i^x \sigma_{i+1}^x + \frac{1-\gamma}{2} \sigma_i^y \sigma_{i+1}^y + \lambda \sigma_i^z \right), \tag{13}$$

where γ is the anisotropic parameter and λ is the strength of the external field. The phase diagram is parametrized by γ and λ. For $\gamma = 0$, the XY model turns into the XX model (isotropic XY model), which has a critical region $\lambda \in [0, 1]$, and the quantum phase transition is of the Berezinskii–Kosterlitz–Thouless type. In this work, we only consider the anisotropic case, $\gamma \in (0, 1]$, where the XY model has a critical point at $\lambda_c = 1$, and the transition belongs to the Ising universality class.

The total spin number is $N = 2M + 1$. This Hamiltonian can be exactly diagonalized by successively applying Jordan–Wigner, Fourier, and Bogoliubov transforms. The final form of the Hamiltonian is given by $H(\gamma, \lambda) = \sum_{k=-M}^{M} \Lambda_k (\hat{b}_k^+ \hat{b}_k - 1)$, with $\Lambda_k = \sqrt{\varepsilon_k^2 + \gamma^2 \sin^2(ka)}$, and $\varepsilon_k = \cos ka - \lambda$, $a = 2\pi/N$. One-particle excitations are created when the fermionic operators $\hat{b}_k = \cos\frac{\theta_k}{2} \hat{d}_k - i \sin\frac{\theta_k}{2} \hat{d}_{-k}^+$ act on the ground state

$$|g(\gamma, \lambda)\rangle = \prod_{k=1}^{M} \left(\cos\frac{\theta_k}{2} |0\rangle_k |0\rangle_{-k} + i \sin\frac{\theta_k}{2} |1\rangle_k |1\rangle_{-k} \right) \tag{14}$$

with $\hat{d}_k |0\rangle_k |0\rangle_{-k} = \hat{d}_{-k} |0\rangle_k |0\rangle_{-k} = \hat{b}_k |g(\gamma, \lambda)\rangle = 0$ and $\cos\theta_k = \varepsilon_k / \Lambda_k$. The phase transition that occurs at $\lambda = 1$ is due to the competition between the neighboring spin-exchanging interactions and the transverse field. The ground state is fully polarized when $\lambda \to \infty$, and is degenerated when $\lambda = 0$.

3.2. Simplified maximal QFI and spin squeezing parameters

By exploring the symmetries of the XY model, the maximal QFI and spin squeezing parameters in this paper can be further simplified. The XY model is of spin-flip symmetry,

$$\left[\prod_{i=-M}^{M} \sigma_i^z, H\right] = 0, \tag{15}$$

which gives

$$\langle S_\alpha \rangle = \langle S_\alpha S_z \rangle = \langle S_z S_\alpha \rangle = 0, \quad \alpha = x, y. \tag{16}$$

Furthermore, since the Hamiltonian is real, the ground state can always be chosen to be real, and thus we have $\langle [S_x, S_y]_+ \rangle = 0$. Therefore, the mean-spin direction is in the z-axis, and there is no correlation between the operators S_α, $(\alpha \in x, y, z)$. The minimal χ^2 is

$$\chi^2 = \frac{N}{\max F_Q} = \frac{N}{4 \max\left(\langle S_x^2\rangle, \langle S_y^2\rangle, \Delta S_z^2\right)}, \tag{17}$$

while the explicit expression for ξ_S^2 is

$$\xi_S^2 = \frac{4 \min\left(\langle S_x^2\rangle, \langle S_y^2\rangle\right)}{N}. \tag{18}$$

As shown above, with the symmetries of the XY model, both QFI and spin squeezing parameters can be expressed in a very simple form.

3.3. Maximal QFI and spin squeezing in the quantum phase transition of the XY model

Now, according to equations (17) and (18), we need to calculate the expectation values of S_α and S_α^2 in the ground state. As presented in [42–44], we have

$$\begin{aligned}
\langle S_z \rangle &= \frac{1}{2}\sum_{i=-M}^{M} \langle \sigma_i^z \rangle = \sum_{k=-M}^{M} \frac{[\lambda - \cos(ka)]}{\Lambda_k}, \\
\langle S_\alpha^2 \rangle &= \frac{N}{4}\sum_{r=0}^{N-1} \langle \sigma_{-M}^\alpha \sigma_{-M+r}^\alpha \rangle,
\end{aligned} \tag{19}$$

where S_α^2 involve correlation terms

$$\begin{aligned}
\langle \sigma_0^x \sigma_r^x \rangle &= \begin{vmatrix} a_{-1} & a_{-2} & \cdots & a_{-r} \\ a_0 & a_{-1} & \cdots & a_{-r+1} \\ \vdots & \vdots & \vdots & \vdots \\ a_{r-2} & a_{r-3} & \cdots & a_{-1} \end{vmatrix}, \\
\langle \sigma_0^y \sigma_r^y \rangle &= \begin{vmatrix} a_1 & a_0 & \cdots & a_{-r+2} \\ a_2 & a_1 & \cdots & a_{-r+3} \\ \vdots & \vdots & \vdots & \vdots \\ a_r & a_{r-1} & \cdots & a_1 \end{vmatrix},
\end{aligned} \tag{20}$$

and

$$a_r = \sum_{k=-M}^{M} \frac{\cos(kra)[\cos(ka) - \lambda] - \gamma \sin(kra)\sin(ka)}{N\Lambda_k}. \tag{21}$$

Substituting the above equations into equations (17) and (18), we can obtain χ^2 and ξ_S^2. In figures 1 and 2, we show numerical results of χ^2 and ξ_S^2 as functions of λ for different γ in

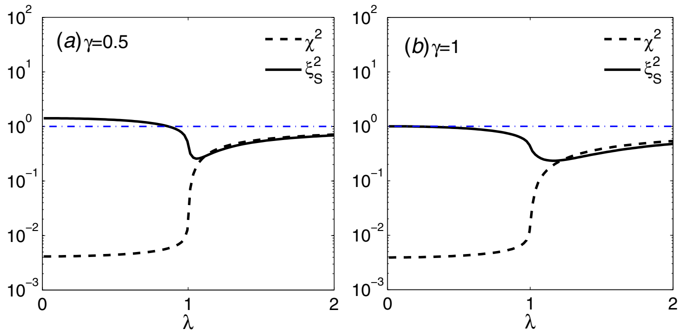

Figure 1. ξ_S^2 and χ^2 as functions of λ for $\gamma = 0.5, 1$ with system size $N = 2^8$. The location of the crossing point of ξ_S^2 and the horizontal dashed line lies at $\lambda = \sqrt{1-\gamma^2}$, where $\xi_S^2 = 1$.

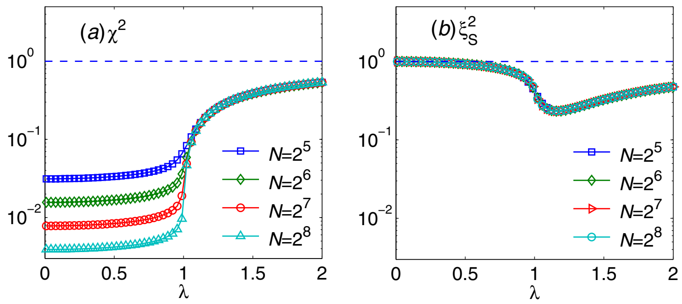

Figure 2. ξ_S^2 and χ^2 as functions of λ for different system sizes $N = 2^5, 2^6, 2^7, 2^8$ with $\gamma = 1$. In (*a*), $\chi^2 \propto 1/N$ in the broken phase and is nearly independent of N in the symmetric phase. In (*b*), the squeezing parameter ξ_S^2 is nearly independent of N in both broken and symmetric phases.

the finite size case. Firstly, we can see that ξ_S^2 is nearly independent of N in both phases. In contrast with ξ_S^2, we note that χ^2 behaves differently in the two phases. It is nearly independent of N in the symmetric phase ($\lambda > 1$), while it scales as $\chi^2 \propto 1/N$ in the symmetry-broken phase ($\lambda < 1$). The results imply that the phase sensitivity of the ground state can be enhanced to the Heisenberg limit if $\lambda < 1$. The scaling behavior of χ^2 for $\lambda = 1/2$ is show in figure 3. From equations (4) and (8) we find that the bound $\Delta\theta_{\mathrm{QCR}}$ approaches the Heisenberg limit, while the influence of γ is small.

Furthermore, as shown in figure 1, when $\lambda < \sqrt{1-\gamma^2}$ the ground state is entangled according to the criterion $\chi^2 < 1$, while not spin squeezed since $\xi_S^2 > 1$. Thus, the spin squeezing parameter fails in detecting the ground state entanglement of the *XY* model.

Now, we study the critical behaviors of χ^2 and ξ_S^2 in the vicinity of the critical point. As shown in figure 2, χ^2 and ξ_S^2 have no singularities, while their derivatives with respect to λ near the critical point tend to be divergent. We perform numerical computing and find that

$$\begin{aligned} \mathrm{d}\xi_S^2/\mathrm{d}\lambda|_{\lambda_m} &\propto -N^{A_1} \\ \mathrm{d}\chi^2/\mathrm{d}\lambda|_{\lambda_m} &\propto N^{A_2}, \end{aligned} \tag{22}$$

where $A_{1,2}$ are constants, λ_m is the position of the maximal derivative. The numerical results are shown in figure 4. Therefore, both χ^2 and ξ_S^2 can well indicate the critical point. Compared with the LMG model, it has been found that the squeezing parameter $\xi_S^2|_{\lambda_m} \propto N^{-0.33\pm0.01}$ [33].

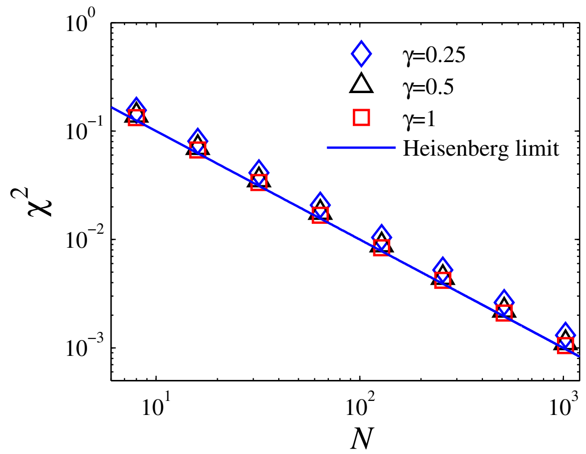

Figure 3. Scaling χ^2 as functions of N for different parameter γ. Set $\lambda = 1/2$. The solid line is the Heisenberg limit $1/N$.

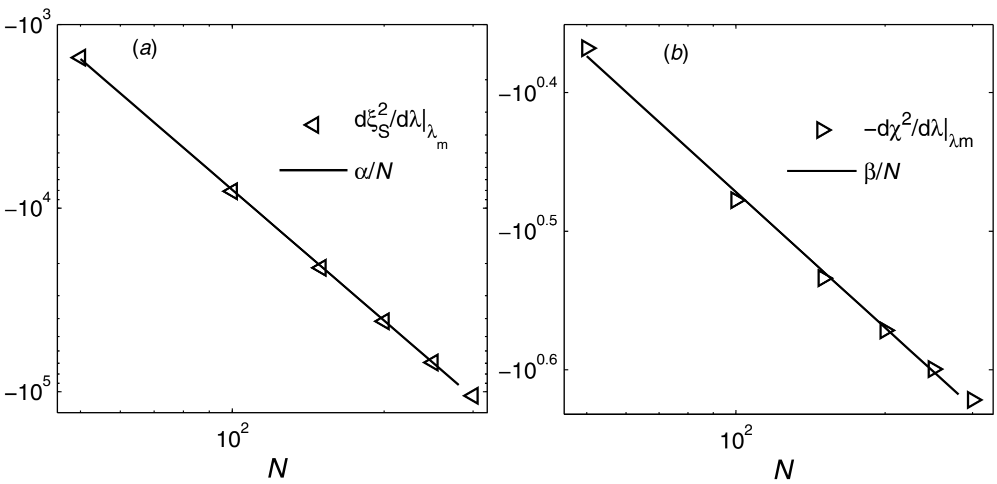

Figure 4. Scaling of $\mathrm{d}\xi_S^2/\mathrm{d}\lambda|_{\lambda_m}$ and $-\mathrm{d}\chi^2/\mathrm{d}\lambda|_{\lambda_m}$ as functions of N for $\gamma = 1/2$. The solid line is a linear fit. In (*a*) and (*b*), $\mathrm{d}\xi_S^2/\mathrm{d}\lambda|_{\lambda_m} \propto -N^{A_1}$ and $-\mathrm{d}\chi^2/\mathrm{d}\lambda|_{\lambda_m} \propto -N^{A_2}$, respectively.

The different behaviors of QFI in the two phases are very interesting, self-consistent results were obtained in another second-order quantum phase transition model, the LMG model [23]. Hence, we believe that the results can be explained by considering the symmetry-broken mechanism. If $\lambda > 1$, the system tends to polarize along the z-direction. Consider a limiting case, $\lambda \to \infty$, the ground state is a coherent spin state $|\psi_g\rangle = |\uparrow\rangle^{\otimes N}$, which is fully polarized in the z-direction. Therefore, the ability of the state in phase estimation process is the same as a classical state. As shown in figure 1, χ^2 tends to 1 with the increase of λ.

On the other hand, if $\lambda < 1$, the spin-exchanging terms are dominant. For the sake of simplicity, we consider the symmetry-broken mechanism in the thermodynamic limit. In this situation, the probabilities of the system that polarize to $\pm x$ directions are equal, and thus the variance of the angular momentum in the x-direction is very large. According to the definition (6), the maximal QFI is proportional to the maximal variance of the system. Therefore, as shown in figure 1, $\chi^2 \propto 1/N$ in the symmetry-broken phase.

4. Conclusion

In summary, we have studied the QFI of the ground state of the XY spin chain. We find that the parameter χ^2 is nearly independent of N in the symmetric phase, and the phase estimation precision is not enhanced essentially. However, in the broken phase $\chi^2 \propto 1/N$, and the phase sensitivity is enhanced to the Heisenberg limit. Furthermore, χ^2 is more effective than ξ_S^2 in the study of the ground-state entanglement for the XY model. We observe a parameter region that the entanglement can be detect by QFI but not spin squeezing. We also study the critical behaviors of ξ_S^2 and χ^2. Their first-order derivatives with respect to λ are singular around the critical point. The scaling form is obtained by numerical computing.

Finally, based on the symmetry broken mechanism, we give an explanation for the different behaviors of χ^2 in the two phases, and this analysis may be extended to a class of models that have second-order phase transition.

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